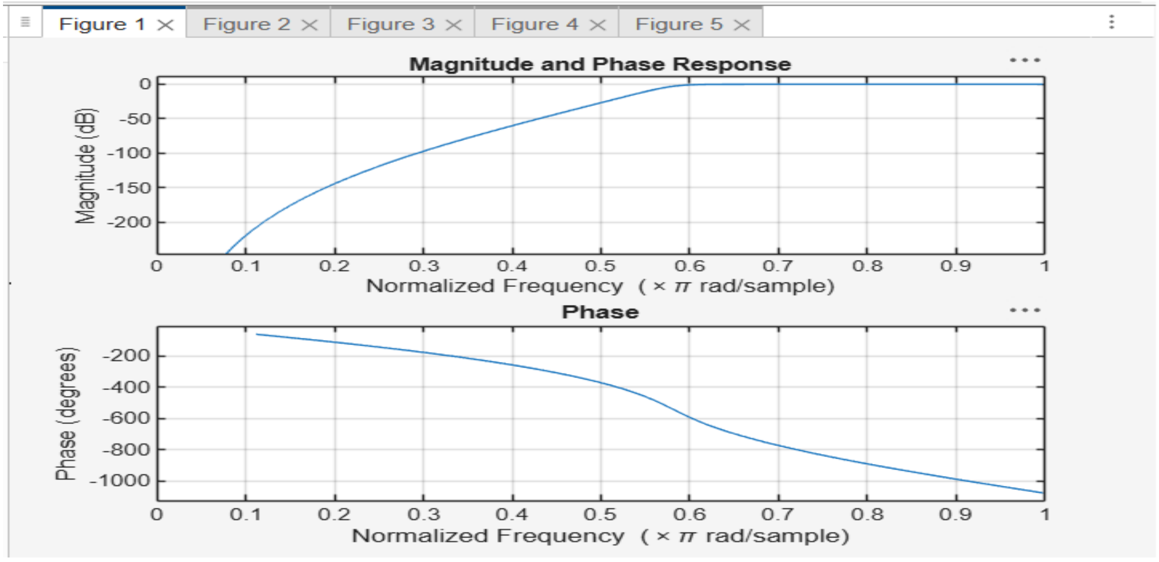
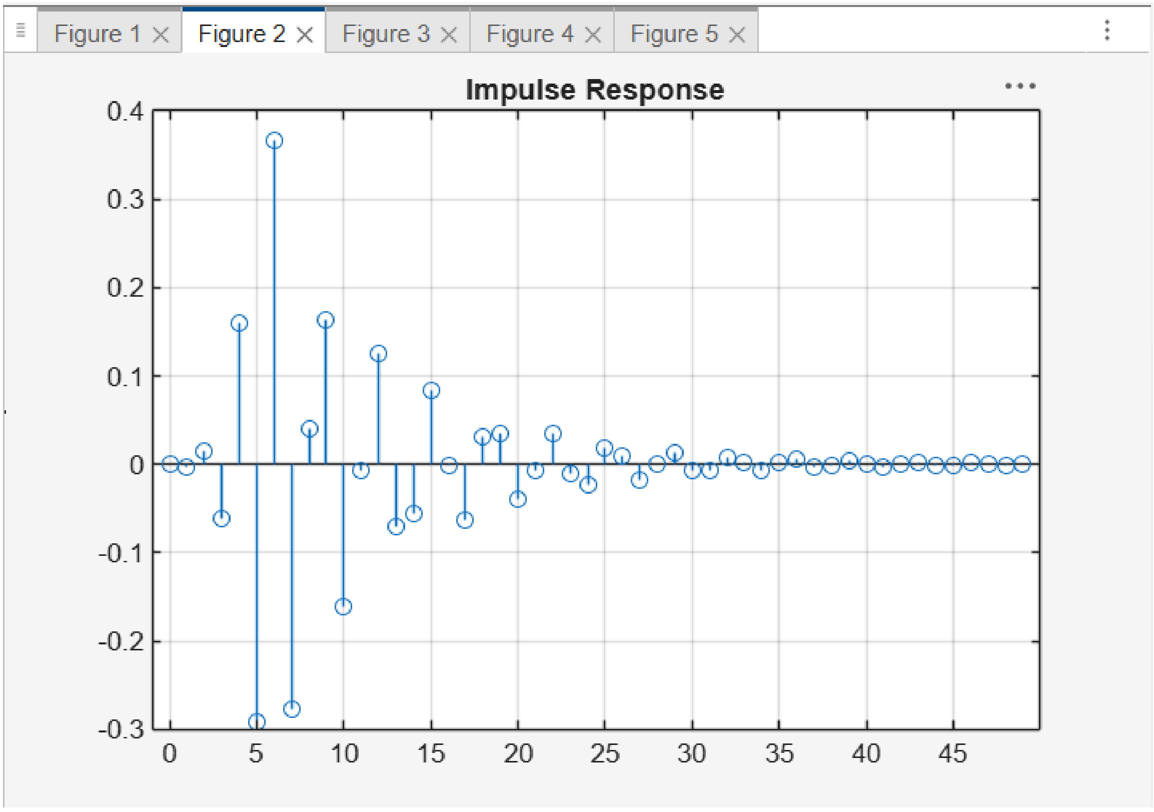
1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

```
clc;
clear;
close all;
% EX NO:1
% Butterworth IIR High-Pass Filter
Fs = 10000;
Fp = 3000;
Fst = 2000;
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a] = butter(N,Wn,'high');
% Magnitude & Phase Response
figure
freqz(b,a)
title('Magnitude and Phase Response')
% Impulse Response
figure
[h,n] = impz(b,a,50);
stem(n,h)
grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
```

```
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```





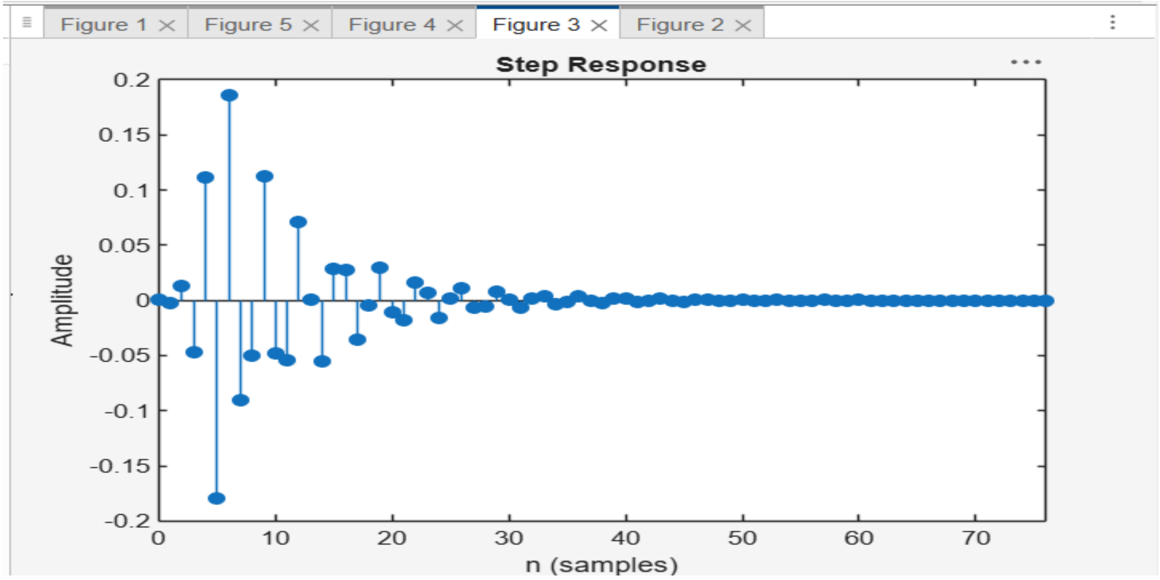


Figure 1
Figure 5
Figure 4
Figure 3
Figure 2
Step Response
Amplitude
n (samples)
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.2
0
10
20
30
40
50
60
70

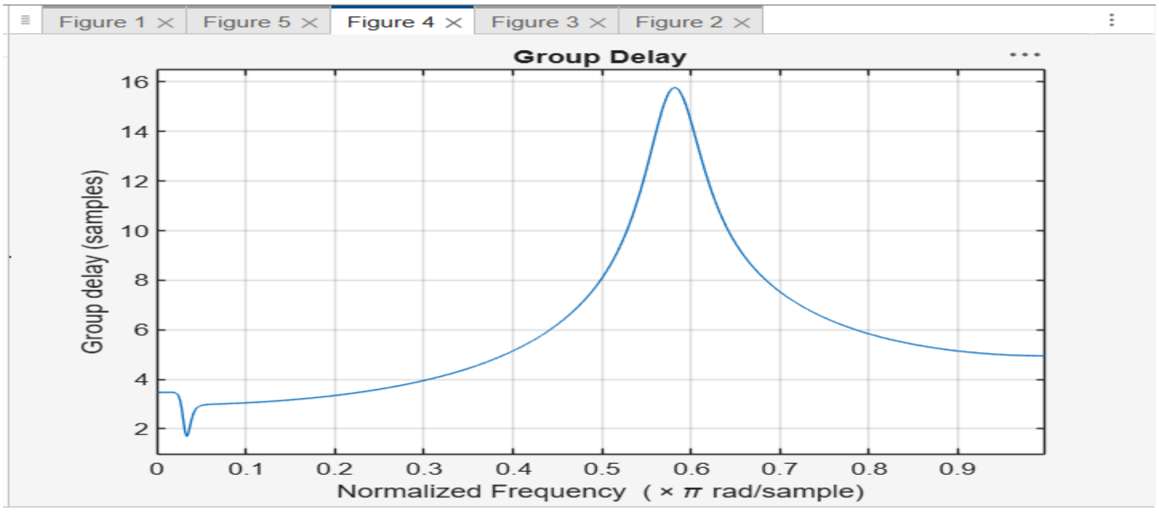


Figure 1
Figure 5
Figure 4
Figure 3
Figure 2
Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)
16
14
12
10
8
6
4
2
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9

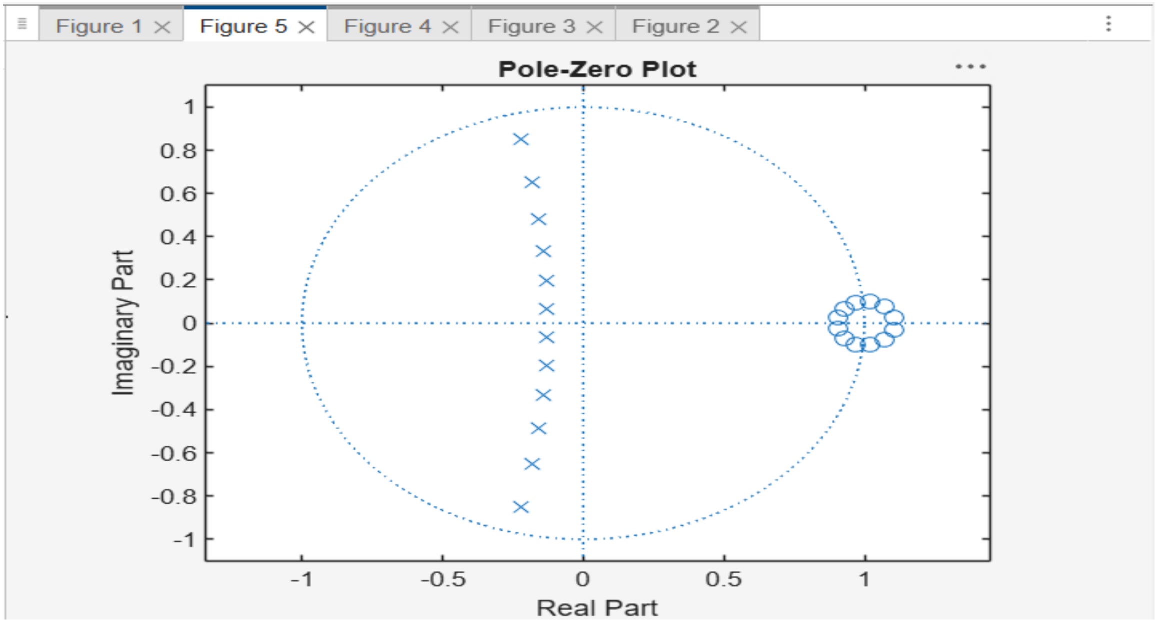


Figure 1
Figure 5
Figure 4
Figure 3
Figure 2
Pole-Zero Plot
Imaginary Part
Real Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1